Abdul Majid Ridwan Tyastonoatmaja

Full-Stack AI Engineer — Business Systems, ML & Automation
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Summary

Full-stack engineer (S.Kom. 2026, GPA 3.99) and sole developer at Ferswit Indonesia, extending the inherited warehouse system used by ~20 staff and running the sales, marketing and automation services around it. Builds ML end to end in research and competition work: a hybrid-RAG thesis (ARCADIA 2026, 2nd place) and SweetLife (AMICTA 2025 AI winner).

Experience

Full-Stack Systems Engineer (AI & Automation), Ferswit Indonesia – Tulungagung May 2026 – present

- Took over the existing company WMS (Node.js/Express, React, PostgreSQL) used by ~20 staff across 3 warehouses and added inter-entity stock transfers with debt settlement, a searchable purchase-order list and a point-of-sale module.
- Built the marketing and reseller-management modules as a Django + PostgreSQL service, both in daily use; set up ERPNext HRMS for HR records.
- Replaced three manual routines with Celery jobs (daily off-site database backups, marketing-record sync to Google Sheets via the Sheets API, TikTok video-stats collection), saving an estimated 30 minutes of manual work a day.
- Deployed and operated every service alone on the production VPS as Docker containers.

AI Engineer (Part-time), Informatics Study Program, Universitas AMIKOM – Yogyakarta Nov 2024 – Apr 2025

- Built a RAG chatbot on Pinecone vector search, and an embedding-based clustering of thesis titles by theme that surfaced trending topics and recurring research objects.

Projects

Hybrid-RAG Tafsir Assistant (Thesis, ARCADIA 2026 2nd Place) – Universitas AMIKOM Sep 2025 – Jan 2026

- Indexed 1,896 passages of Tafsir Ibn Kathir (6,500+ pages) for hybrid BM25 + E5-large multilingual retrieval; the hybrid retriever scored nDCG@10 0.94 (lexical relevance) and 0.99 (semantic relevance) on 154 queries auto-extracted from the corpus.
- Grounded LLaMA 3.2 3B answers (served via Hugging Face API) in cited passages; 0.972 context relevance (embedding cosine to the source) and 5.93 s average response, with retrieval on a CPU-only 4 GB machine.

IJO PLIS: IHSG & USD/IDR Forecasting Platform – Personal May 2026 – Jun 2026

- Built a Django + PostgreSQL app that forecasts the Indonesia Composite Index and USD/IDR 7, 30 and 90 days ahead with Prophet on ~10 years of data, with trend and seasonality decomposition.
- Automated weekly retraining (GitHub Actions cron, Celery Beat), versioning every model and promoting the lowest-MAE one; 9 unit tests, 6 JSON API endpoints, Docker.
- Backtested over 11 rolling origins (2024 – 2026): Prophet did not beat a naive last-value baseline (90-day MAPE 7.4% vs 5.0% on IHSG, 1.8% vs 1.7% on USD/IDR); the dashboard shows forecasts as confidence bands, not price targets.

NyatetDuwit: Offline-First Finance PWA – Personal May 2026 – present

- Built an installable PWA that logs an expense in under 3 seconds with no account or server — IndexedDB storage, budgets, recurring entries, reminders and JSON backup.

Education

Universitas AMIKOM Yogyakarta, S.Kom. in Informatics, GPA 3.99/4.00 – Yogyakarta 2022 – 2026

- Laboratory Assistant (part-time, Sep 2024 – Jan 2026): assisted practical sessions and administered exams in Python, data science, Tableau and SPSS

Awards

- Winner, AI category, AMICTA 2025: SweetLife, a team-built diabetes-care app (Bangkit ML capstone) with nutrition detection from food photos, diabetes-risk prediction, an LLM chatbot and meal and exercise recommendations
- 2nd Place, AI, ARCADIA Award 2026: the Hybrid-RAG Tafsir Assistant (thesis above)
- 2nd Place, IoT, Ignition Informatics 2024: Smart Laundry Bin, an ESP8266 load and detergent monitor

Skills

Backend & data: Python, Django, Node.js / Express, PostgreSQL, SQL, Celery + Redis, REST APIs

Frontend: React, JavaScript, Tailwind CSS, Chart.js, PWA / IndexedDB

AI / ML: RAG, LLMs, embeddings / vector search, TensorFlow, Prophet / time series, clustering

Ops & tools: Docker, Linux / VPS, Git, GitHub Actions, ERPNext, Google Sheets API, Pinecone

Languages: Indonesian (native), English (professional, EF SET C2 Proficient)